

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1-2

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1. TITLE

Create an Interactive Styled Webpage Using HTML5 Canvas and CSS3

2. AIM

To design an attractive webpage using HTML5 and CSS3 and demonstrate canvas graphics, navigation menus, Flexbox layouts, cards, buttons, gradients, borders, shadows, rounded corners and footer styling.

3. OBJECTIVE

The objectives of this experiment are:

- To create a structured webpage using HTML5.
 - To apply CSS3 properties for improving webpage appearance.
 - To design a navigation bar with styled links.
 - To use CSS gradients and text-shadow effects, create graphical elements using HTML5 Canvas.
 - To arrange three information cards using CSS Flexbox.
 - To demonstrate styled buttons and hover effects, display graphical shapes and text using JavaScript canvas methods.
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4. THEORY

HTML5 provides several elements that help create modern webpages. One such feature is the Canvas element, which provides a drawing area that can be controlled using JavaScript. CSS3 is used to improve the visual appearance of a webpage. It provides properties for colors, gradients, spacing, borders, rounded corners, shadows, transitions and hover effects. In this experiment, the webpage contains a header, navigation bar, canvas area, three information cards and a footer. CSS Flexbox is used to arrange the cards horizontally and allow them to wrap when required. JavaScript accesses the canvas using `getElementById()` and obtains the two-dimensional drawing context using `getContext("2d")`. Different canvas methods are then used to create a rectangle, circle, line and text. The webpage therefore combines HTML5 for structure, CSS3 for presentation and JavaScript for graphical drawing.

5. ALGORITHM / PROCEDURE

1. Create the HTML document with the required metadata and title.
 2. Add a header containing the webpage title and description.
 3. Create a navigation bar containing multiple navigation links.
 4. Add a canvas element with a specified width and height.
 5. Apply CSS styling to the canvas using borders, background and shadows.
 6. Create a Flexbox container for the information cards.
 7. Add three cards with headings, descriptions and buttons.
 8. Apply CSS properties such as padding, margins, borders, radius and shadows.
 9. Define hover effects for navigation links and buttons.
 10. Create a footer at the bottom of the webpage. Access the canvas using JavaScript.
 12. Obtain the 2D drawing context. Draw a rectangle, circle and line using canvas methods.
 14. Display text inside the canvas. Open the HTML file in a browser and verify the complete webpage.
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6. PROGRAM

code.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>

<style>
/* ===== General Styling ===== */
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
}

body{
    font-family:Arial, sans-serif;
    background:linear-gradient(to right,#74ebd5,#9face6);
    color:#333;
}

/* ===== Header ===== */
header{
    background:#2c3e50;
    color:white;
    text-align:center;
    padding:20px;
    text-shadow:2px 2px 5px black;
}

/* ===== Navigation ===== */
nav{
    background:#34495e;
    padding:15px;
    text-align:center;
}

nav a{
    color:white;
    text-decoration:none;
```

```

        margin:20px;
        font-size:18px;
        font-weight:bold;
        transition:0.3s;
    }

    nav a:hover{
        color:yellow;
        background:#1abc9c;
        padding:8px 15px;
        border-radius:5px;
    }

    /* ===== Canvas Section ===== */
    .canvas-section{
        text-align:center;
        margin:30px;
    }

    canvas{
        border:3px solid black;
        background:white;
        box-shadow:5px 5px 10px gray;
    }

    /* ===== Cards ===== */
    .container{
        display:flex;
        justify-content:center;
        gap:20px;
        flex-wrap:wrap;
        margin:30px;
    }

    .card{
        width:280px;
        background:white;
        padding:20px;
        border:2px solid #3498db;
    }

    .card h3{
        font-family:Georgia;
        color:#2c3e50;
    }

    .card p{
        margin-top:10px;
        font-size:16px;
    }

    button{
        margin-top:15px;
        padding:10px 20px;
        border:none;
        border-radius:10px;
        background:#3498db;
        color:white;
        font-size:16px;
        cursor:pointer;
        transition:0.3s;
    }

    button:hover{

```

```

        background:#2ecc71;
        transform:scale(1.1);
    }

/* ===== Footer ===== */
footer{
    background:#2c3e50;
    color:white;
    text-align:center;
    padding:15px;
    margin-top:20px;
    text-shadow:2px 2px 5px black;
}
</style>

</head>
<body>

<header>
    <h1>My Styled Webpage</h1>
    <p>HTML5 Canvas and CSS3 Styling Demo</p>
</header>

<nav>
    <a href="#">Home</a>
    <a href="#">About</a>
    <a href="#">Services</a>
    <a href="#">Contact</a>
</nav>

<section class="canvas-section">
    <h2>HTML5 Canvas</h2>
    <br>
    <canvas id="myCanvas" width="600" height="300"></canvas>
</section>

<section class="container">

<div class="card">
<h3>Card One</h3>
<p>This card demonstrates borders, padding, margins, rounded corners and shadow effects.</p>
<button>Read More</button>
</div>

<div class="card">
<h3>Card Two</h3>
<p>CSS3 makes webpages attractive using gradients, hover effects, fonts and colors.</p>
<button>Learn More</button>
</div>

<div class="card">
<h3>Card Three</h3>
<p>HTML5 Canvas allows drawing graphics directly inside a webpage using JavaScript.</p>
<button>Explore</button>
</div>

</section>

<footer>
    © 2026 My Styled Webpage | All Rights Reserved
</footer>

<script>

```

```

var canvas = document.getElementById("myCanvas");
var ctx = canvas.getContext("2d");

/* Rectangle */
ctx.fillStyle="blue";
ctx.fillRect(40,40,150,80);

/* Circle */
ctx.beginPath();
ctx.arc(300,80,40,0,2*Math.PI);
ctx.fillStyle="red";
ctx.fill();

/* Line */
ctx.beginPath();
ctx.moveTo(420,40);
ctx.lineTo(550,150);
ctx.strokeStyle="green";
ctx.lineWidth=4;
ctx.stroke();

/* Name Text */
ctx.font="28px Arial";
ctx.fillStyle="purple";
ctx.fillText("Hansikha",220,220);

</script>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
body	Sets the Arial font, gradient background and text color.
header	Styles the heading area with dark background, white text and shadow.
Code	Explanation
nav / nav a	Creates the navigation bar and styled navigation links.
.canvas-section	Centers the canvas section and adds spacing.
canvas	Adds a black border, white background and shadow to the canvas.
.container / .card	Arranges and styles the three cards using Flexbox.
button / button:hover	Styles the buttons and changes their appearance on hover.
getContext("2d")	Obtains the 2D drawing context for the canvas.
fillRect() / arc()	Draws the blue rectangle and red circle.
moveTo() / lineTo() / stroke()	Draws the green line.
fillText()	Displays the text “vijayasri” on the canvas.
footer	Creates the styled footer at the bottom of the page.

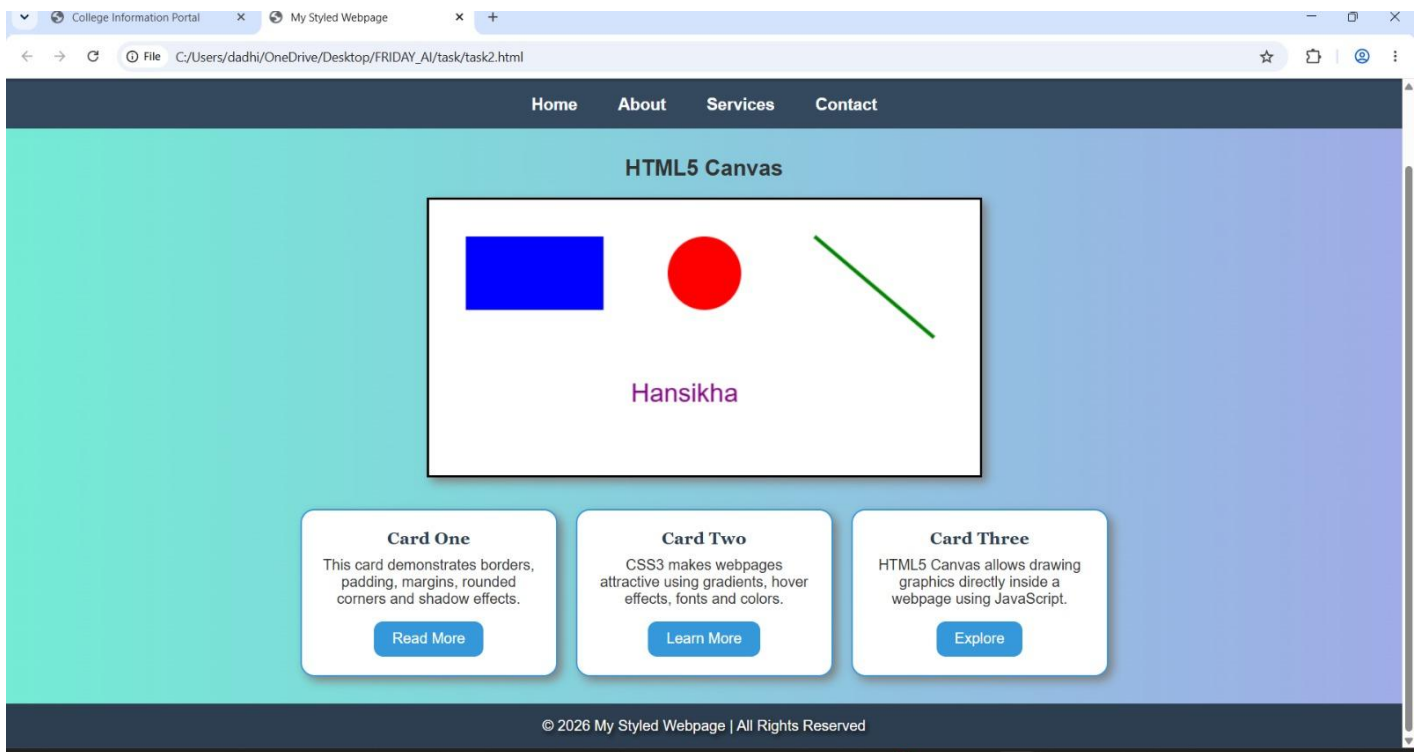
8. TEST CASES AND EXECUTION DATA

The following observations are based on the provided repository output screenshots. No unshown execution result has been invented.

Test Case	Input / Feature	Expected Result	Actual Result	Status
TC-01	Canvas drawing	Rectangle, circle, line and text should appear	All four canvas drawings are visible in the supplied output	Observed
TC-02	Three cards	Cards should be arranged and styled	Three bordered, rounded cards with shadows are visible	Observed
TC-03	Buttons	Each card should contain a styled button	Read More, Learn More and Explore buttons are visible	Observed
TC-04	Page styling	Gradient background, navigation and footer should render	The supplied output shows the gradient page, navigation and footer	Observed

9. OUTPUT

Output 1: HTML5 Canvas, styled cards and footer



10. RESULT

Thus, an interactive styled webpage was developed using HTML5, CSS3 and JavaScript. The webpage successfully demonstrates gradient styling, navigation, canvas graphics, Flexbox-based cards, buttons, hover effects, shadows, borders and footer design.
